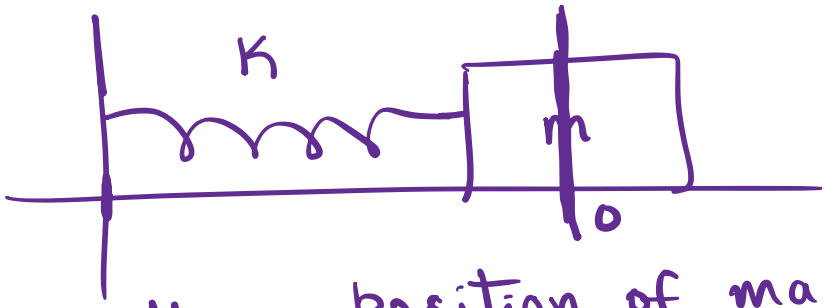
Lecture 22 (153-248)

VAR (Vector Auto Regression)

Revisit Univariate AR



y_t : position of mass at time t

$t = 1, 2, \ldots$

Δ : gap between successive observations

relative of rest position

Time t : Force at time t = mass $\times$ acceleration at time t

y_{t-1}

$-K y_{t-1}$ → force due to spring

friction coefficient → $-\gamma \left(\frac{y_{t-1} - y_{t-2}}{\Delta} \right)$

Total force at time t :

$$-K y_{t-1} - \gamma \frac{y_{t-1} - y_{t-2}}{\Delta}$$

$$= m \times \frac{(y_t - y_{t-1}) - (y_{t-1} - y_{t-2})}{\Delta^2}$$

$$y_t - 2y_{t-1} + y_{t-2} = -\frac{k\Delta^2}{m} y_{t-1} - \frac{\gamma\Delta}{m}(y_{t-1} - y_{t-2})$$

$$y_t = \left(2 - \frac{k\Delta^2}{m} - \frac{\gamma\Delta}{m}\right) y_{t-1} - y_{t-2}\left(1 - \frac{\gamma\Delta}{m}\right)$$

$$a = \frac{k\Delta^2}{m}, \quad g = \frac{\gamma\Delta}{m}$$

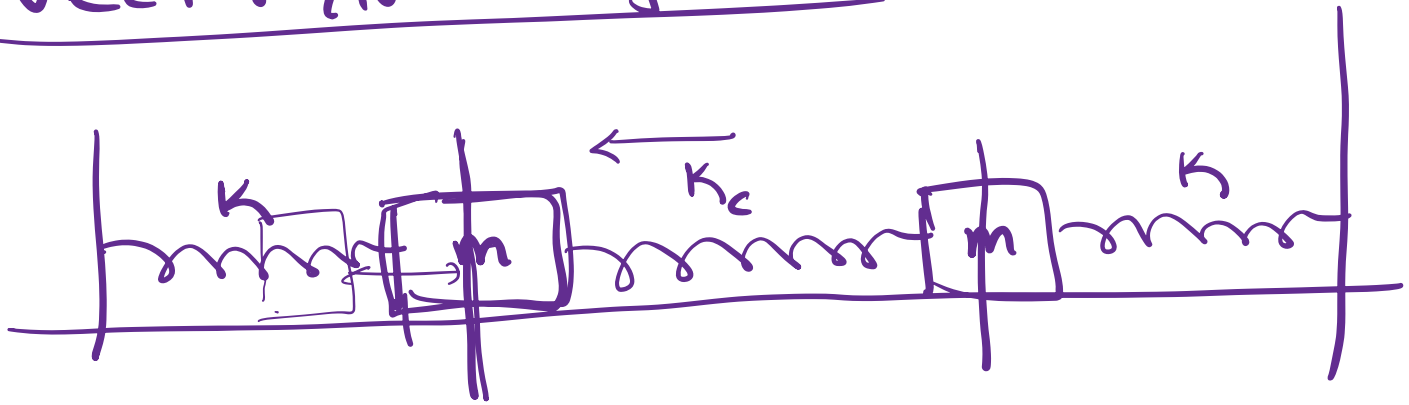
$$y_t = (2 - a - g) y_{t-1} - (1 - g) y_{t-2}$$

$$y_t = (2 - a - g) y_{t-1} - (1 - g) y_{t-2} + \varepsilon_t$$

$$\varepsilon_t \overset{iid}{\sim} N(0, \sigma^2)$$

Example of AR(2)

Vector Auto Regression



$$y_t = \begin{pmatrix} y_t^{(1)} \\ y_t^{(2)} \end{pmatrix}$$

$y_t^{(1)}$: position of body 1 at time t

$y_t^{(2)}$: position of body 2 at time t

Differential Equations:

Body 1

Force at time t:

$$-K y_{t-1}^{(1)} - K_c\left(y_{t-1}^{(1)} - y_{t-1}^{(2)}\right) - \gamma\left[\frac{y_{t-1}^{(1)} - y_{t-2}^{(1)}}{\Delta}\right] = m\,\frac{y_t^{(1)} - 2y_{t-1}^{(1)} + y_{t-2}^{(1)}}{\Delta^2}$$

$$y_t^{(1)} = \left(2 - \frac{K\Delta^2}{m} - \frac{K_c\Delta^2}{m} - \frac{\gamma\Delta}{m}\right) y_{t-1}^{(1)} + \frac{K_c\Delta^2}{m} y_{t-1}^{(2)} - \left(1 - \frac{\gamma\Delta}{m}\right) y_{t-2}^{(1)}$$

$$a = \frac{K\Delta^2}{m}, \quad b = \frac{K_c\Delta^2}{m}, \quad g = \frac{\gamma\Delta}{m}$$

$$y_t^{(1)} = (2 - a - b - g)\, y_{t-1}^{(1)} + b\, y_{t-1}^{(2)} - (1-g)\, y_{t-2}^{(1)}$$

$$y_t^{(2)} = (2 - a - b - g)\, y_{t-1}^{(2)} + b\, y_{t-1}^{(1)} - (1-g)\, y_{t-2}^{(2)}$$

To construct a model, we add noise terms $\varepsilon_t^{(1)}, \varepsilon_t^{(2)}$

$$y_t^{(1)} = (2-a-b-g)\, y_{t-1}^{(1)} + b\, y_{t-1}^{(2)} - (1-g)\, y_{t-2}^{(1)} + \varepsilon_t^{(1)}$$

$$y_t^{(2)} = (2-a-b-g)\, y_{t-1}^{(2)} + b\, y_{t-1}^{(1)} - (1-g)\, y_{t-2}^{(2)} + \varepsilon_t^{(2)}$$

VAR

$$y_t = \begin{pmatrix} y_t^{(1)} \\ y_t^{(2)} \end{pmatrix}$$

$$y_t = \begin{bmatrix} 2-a-b-g & b \\ b & 2-a-b-g \end{bmatrix} y_{t-1} + \begin{bmatrix} -(1-g) & 0 \\ 0 & -(1-g) \end{bmatrix} y_{t-2} + \varepsilon_t$$

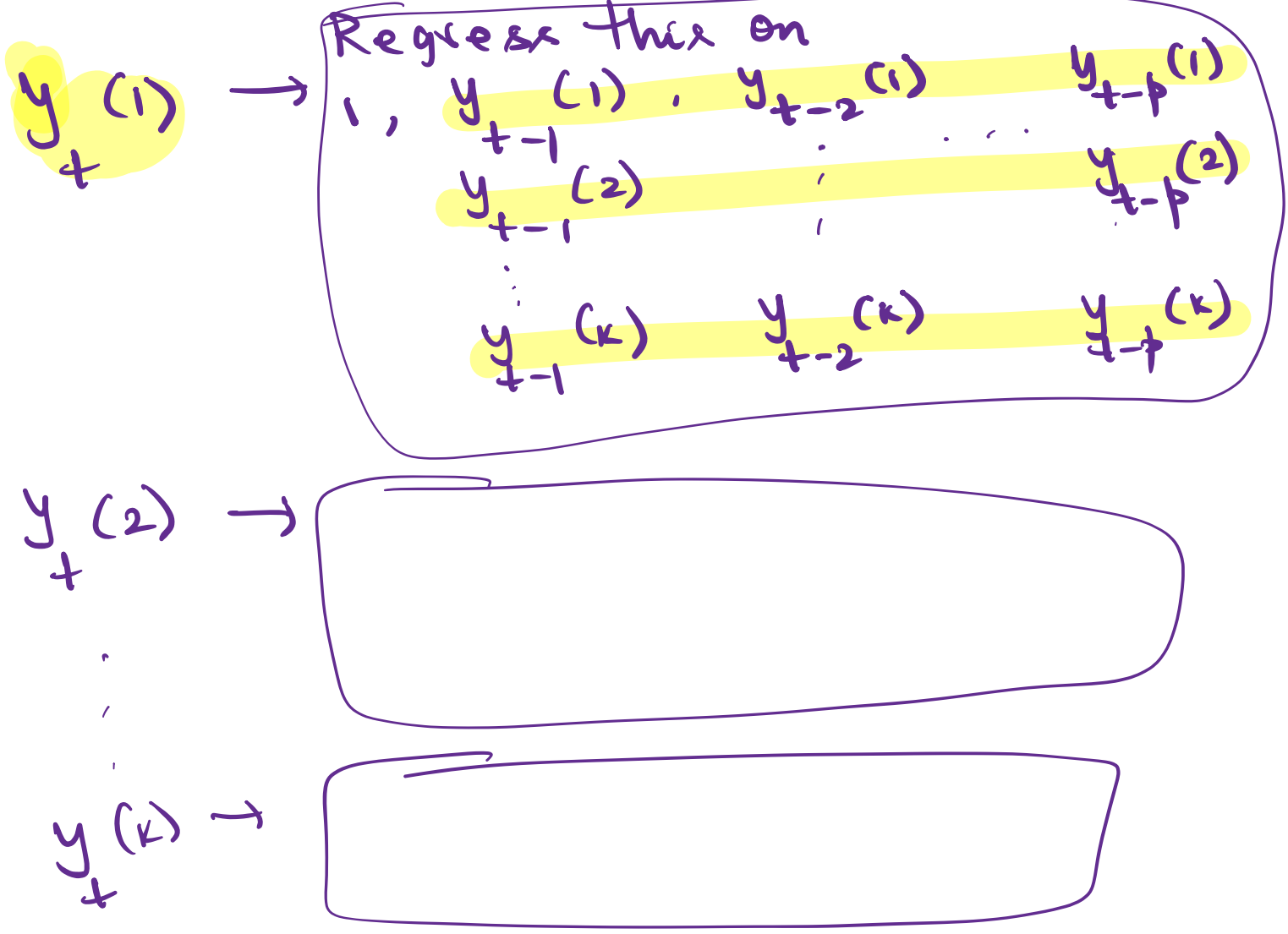
$$y_t = A_1 y_{t-1} + A_2 y_{t-2} + \varepsilon_t$$

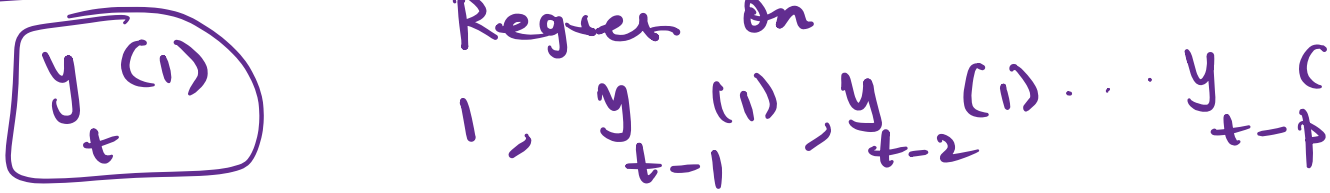
$$y_t = c + A_1 y_{t-1} + A_2 y_{t-2} + \varepsilon_t$$

$\rightarrow$ VAR(2)

VAR(p)

$$y_t = c + A_1 y_{t-1} + \cdots + A_p y_{t-p} + \varepsilon_t$$





AR(p) for $y_t^{(1)}$

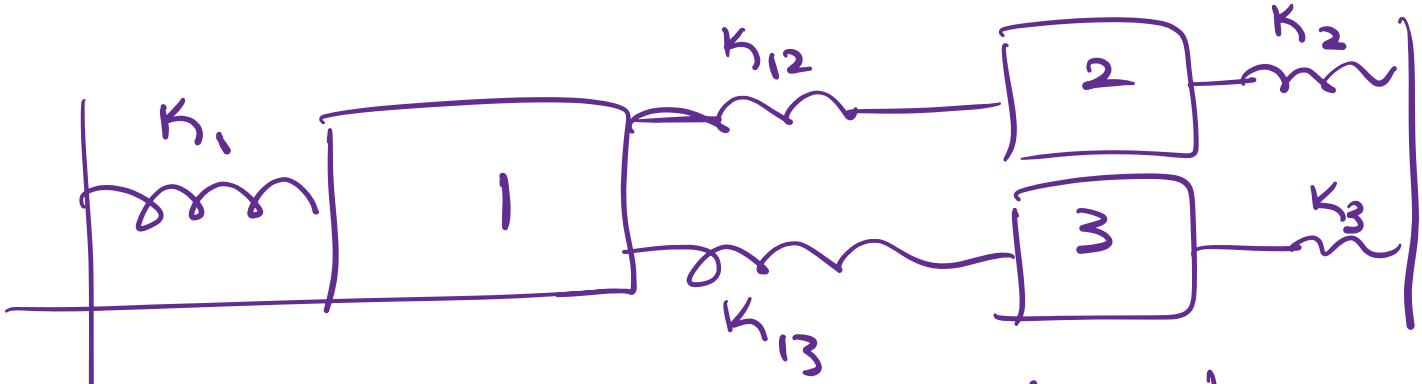
k - Connected Bodies

① Body i has mass m_i & friction coefficient γ_i

② Body i is connected to a wall with a spring with constant k_i.

③ Bodies i & j are connected to each

other with a spring with constant $K_{ij} = K_{ji}$



Check that the model equations become:

$$y_t(i) = \left[2 - \frac{\Delta^2 K_i}{m_i} - \sum_{j:j\neq i} \frac{\Delta^2 K_{ij}}{m_i} - \frac{\Delta\gamma_i}{m_i}\right] y_{t-1}(i)$$

$$+ \sum_{j:j\neq i} \frac{\Delta^2 K_{ij}}{m_i} y_{t-1}(j) - \left(1 - \frac{\Delta\gamma_i}{m_i}\right) y_{t-2}(i)$$

$$+ \varepsilon_t(i)$$

VAR

$$y_t = A_1 y_{t-1} + A_2 y_{t-2} + \varepsilon_t \rightarrow \text{VAR}(2)$$

$A_1 =$ [ ]

$$A_1(i,j) = \frac{\Delta^2 K_{ij}}{m_i} \quad \text{for } i \neq j$$